

Explanatory power: 55 percent

This equation informs us that the sustainable withdrawal rate in retirement can be expected to fall as PE10 is higher and as interest rates are lower. The precise numbers in this equation were estimated from the historical relationship between sustainable spending rates and these variables. We take the natural logarithm of PE10 in order to fit a more precise curve for the relationship between market valuations and spending rates.

Exhibit 5.8 shows the historical sustainable spending rates from 1881 to 1987 using Shiller's data for large-capitalization stocks and ten-year Treasuries. With this data, a hypothetical 1937 retiree saw a historical SAFEMAX of 3.95 percent. This is slightly different from previous results we discussed, as the data set and bond index are different. Every data set is a little different.

The exhibit also provides the estimates developed through the aforementioned equation for the entire time period through the start of 2017. Until 1987, we can compare the relationship between the predictions and the actual values and see it is close. The two variables (PE10 and interest rates) can explain 55 percent of the fluctuations in the historical spending rates. We do not yet know what the thirty-year sustainable spending rates will be after 1987, but we can use this equation to estimate sustainable spending for more recent years, as we know the values of PE10 and the ten-year Treasury rate at the start of those years. We can estimate a sustainable withdrawal rate from any point in time by using the current values for these variables. For instance, in January 2017, PE10 was about 28.2, and the ten-year Treasury was 2.45 percent. Plugging these into the fitted equation gives us:

Predicted Withdrawal Rate for January 2017 Retirees =

$$12.15 - 2.47 \times \log(28.2) + 0.05 \times 2.45 = 4.02\%$$

According to this model, the best guess we can make for the sustainable spending rate for a new retiree in 2017 is 4.02 percent. While this is almost identical to the historical SAFEMAX, it is important to emphasize that this is *not* an estimate of the “safe” withdrawal rate. The actual withdrawal rate is *not* a conservative estimate about a safe withdrawal rate; rather, it is the best